

Analysis of Truss Structures: Manual vs Simulation Approach (May 2025)

Frais Asghar 516900. Shams Manzoor 508560. Muhammad Qasim 512830. Dawood Ahmad 518329.
Muhammad Usman 510379, and Hassan Tariq 523482, *Member, IEEE*

Abstract— In this lab, we are given task of making truss, finding out internal and reaction forces. For this, we used one roller, one pin joint support and applied load at the other joint at which no other reaction force is applied. We used method of joints to find out forces in the truss and also verified the results using ANSYS Workbench. Both the results were much closer with a minute percentage of error which show that our manual calculations are correct. This OEL helps us understanding how to calculate internal and reaction forces of truss and verify results using simulation techniques in ANSYS.

Keywords— Truss analysis, internal forces, reaction forces, method of joints, structural simulation, ANSYS Workbench, static equilibrium, engineering mechanics lab, finite element analysis (FEA), experimental validation.

I. INTRODUCTION

Truss is a triangular structure made up of various members which is used in bridges, roofs and various other structures. We were asked to make a 2D truss with at least 5 members and 3 joints. Then we apply various reaction forces and a load and then find out the external and internal forces of the truss structure. There are 2 methods of solving for these forces; method of sections and method of joints. We are using method of joints because it is more convenient to use in these types of trusses. One force is pinned and other is roller so that our structure doesn't move and stay in equilibrium. After solving manually, we verified our results using ANSYS Workbench. This OEL teaches us real applications of engineering statics in structure solving.

II. THEORY

A. Definition and Assumptions

A truss is a rigid structure composed of straight members connected at joints, designed to carry loads efficiently. The analysis of trusses relies on the following key assumptions:

1. All members are connected by frictionless pins at their ends.
2. Loads are applied only at the joints.
3. Members are subjected to axial forces only (tension or compression).
4. The weight of the members is negligible compared to the applied loads.

B. Method of Joints

The method of joints is a systematic approach to determine internal forces in truss members by analyzing equilibrium at each joint. At every joint, the sum of forces in both horizontal (x) and vertical (y) directions must equal zero.

C. Analysis of Given Truss

External Forces:

The truss in question has the following external forces and reactions:

1. A 300N horizontal force applied at joint D.
2. Vertical reactions at supports A (R_{Ay}) and C (R_{Cy}).
3. A horizontal reaction at support C (R_{Cx}).

Using equilibrium equations:

$\Sigma F_x = 0$: Determined $R_{Cx} = 300\text{N}$ (leftward) to balance the horizontal force at D.

$\Sigma M_C = 0$: Calculated $R_{Ay} = -150\text{N}$ (downward), with a magnitude of 150N.

$\Sigma F_y = 0$: Found $R_{Cy} = 150\text{N}$ (upward) to balance the vertical forces.

Internal Forces:

The internal forces in the truss members were determined using the method of joints:

1. Joint A

- Vertical equilibrium yielded $F_{AD} = 212.13\text{N}$ (tension).
- Horizontal equilibrium gave $F_{AB} = -150\text{N}$ (compression).

2. Joint B:

- Horizontal equilibrium showed $F_{BC} = 150\text{N}$ (tension).
- Vertical equilibrium revealed $F_{BD} = 0\text{N}$ (zero-force member).

3. Joint C:

- Horizontal equilibrium resulted in $F_{CD} = -212.13\text{N}$ (compression).

D. External Forces in the Truss

External forces are the applied loads and support reactions that act on the truss as a whole. In this case, the 300N horizontal force at joint D is the primary external load. The reactions at supports A and C (R_{Ay} , R_{Cy} , and R_{Cx}) are developed to maintain equilibrium. These forces ensure the truss remains stationary under the applied load. The horizontal reaction R_{Cx} balances the 300N force, while the vertical reactions R_{Ay} and R_{Cy} counteract any rotational effects, ensuring moment equilibrium.

E. Internal Forces in the Truss

Internal forces are the axial forces within the truss members, which can be either tensile (pulling the member apart) or compressive (pushing the member together). In this truss:

Members AD and BC are in tension, with forces of 212.13N and 150N, respectively.

Members AB and CD are in compression, with forces of 150N and 212.13N, respectively.

Member BD is a zero-force member, meaning it carries no load under the current configuration. These internal forces are critical for determining the structural integrity of the truss and ensuring each member can withstand the stresses induced by the applied loads.

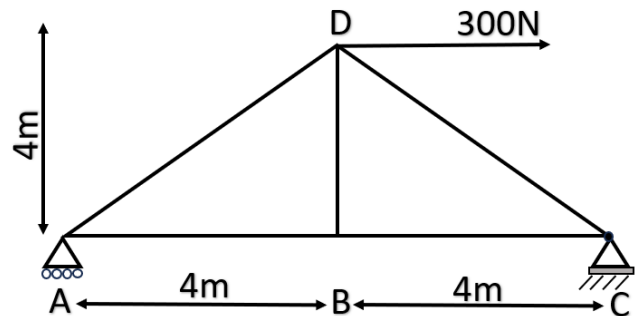
F. Procedure

- Start by observing the truss structure and identifying all joints, members, supports, and applied loads.
- In this case, a 300N horizontal force is applied at joint D. The truss has a roller support at A and a pin support at C.
- Assume the reaction forces at the supports: vertical reactions at A (R_{Ay}) and C (R_{Cy}), and a horizontal reaction at C (R_{Cx}).
- Apply the equilibrium equations to the whole truss:
 - From $\sum F_x = 0$, calculate $R_{Cx} = 300\text{N}$ (leftward) to balance the horizontal force.
 - Take moments about point C ($\sum M_C = 0$) to find $R_{Ay} = -150\text{N}$ (downward).

- Use $\sum F_y = 0$ to calculate $R_{Cy} = 150\text{N}$ (upward), balancing all vertical forces.

- With support reactions known, move to internal forces using the method of joints.
- Start at joint A:
 - Use vertical equilibrium to find $F_{AD} = 212.13\text{N}$ (tension).
 - Use horizontal equilibrium to find $F_{AB} = -150\text{N}$ (compression).
- Move to joint B:
 - Horizontal forces give $F_{BC} = 150\text{N}$ (tension).
 - Vertical equilibrium shows $F_{BD} = 0\text{N}$, making it a zero-force member.
- At joint C:
 - Horizontal equilibrium gives $F_{CD} = -212.13\text{N}$ (compression).
- Conclude the process by checking if all joints are in equilibrium and verifying results through software like ANSYS if needed.

G. Truss and Solution

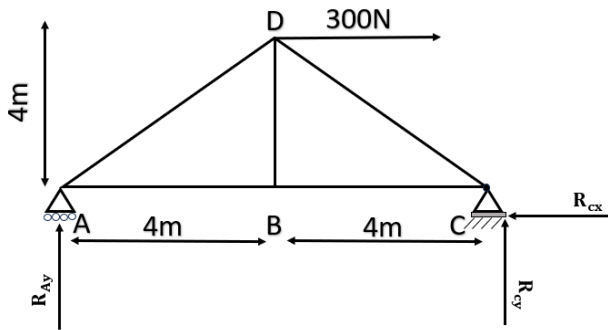


In this task, we designed a simple truss consisting of five joints (A, B, C, D, and one vertical member BD) and six members forming two triangles. The truss is supported with a roller at joint A and a pin at joint C. The geometry of the truss includes 4 meters of horizontal length between each joint and 4 meters of vertical height at joint D. A horizontal load is applied at joint D.

The applied load is calculated using the last two digits of Qasim's CMS ID. Taking the digits as 30, the applied load becomes:

$$\text{Load} = 30 \times 10 = 300\text{ N}$$

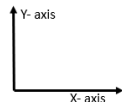
H. Determining External Forces



$$\begin{aligned}\sum F_x &= 0 \\ 300 - R_{Cx} &= 0 \\ R_{Cx} &= 300 \text{ N (Leftward)}\end{aligned}$$

$$\begin{aligned}\sum M_C &= 0 \text{ (Taking clockwise as positive)} \\ (R_{Ay})(8) + (300)(4) &= 0 \\ R_{Ay} &= \frac{-1200}{8} \\ R_{Ay} &= -150 \text{ (Downward)} \\ R_{Ay} &= 150 \text{ N (Magnitude)}\end{aligned}$$

$$\begin{aligned}\sum F_y &= 0 \\ R_{Ay} + R_{Cy} &= 0 \\ R_{Cy} &= 150 \text{ N (Upward)}\end{aligned}$$



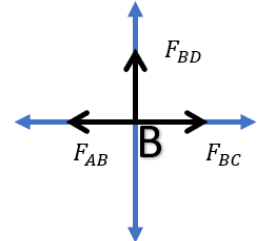
Joint B:

Horizontal:

$$\begin{aligned}F_{BC} - F_{AC} &= 0 \\ F_{BC} &= 150 \text{ N (Tension)}\end{aligned}$$

Vertical:

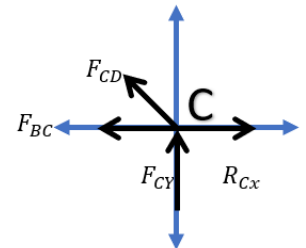
$$F_{BD} = 0 \text{ N (Zero-Force)}$$



Joint C:

Horizontal:

$$\begin{aligned}-F_{BC} + F_{CD} \cos 45^\circ + R_{Cx} &= 0 \\ -150 + F_{CD} \cos 45^\circ + 300 &= 0 \\ F_{CD} &= -212.13 \text{ N (Compression)}\end{aligned}$$



I. Use Method of Joints to Find Internal Forces

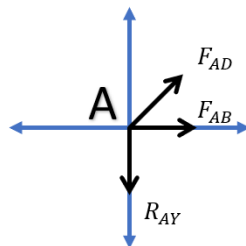
Joint A:

Vertical:

$$\begin{aligned}F_{AD} \sin 45^\circ - 150 &= 0 \\ F_{AD} \sin 45^\circ &= 150 \\ F_{AD} &= 212.13 \text{ N (Tension)}\end{aligned}$$

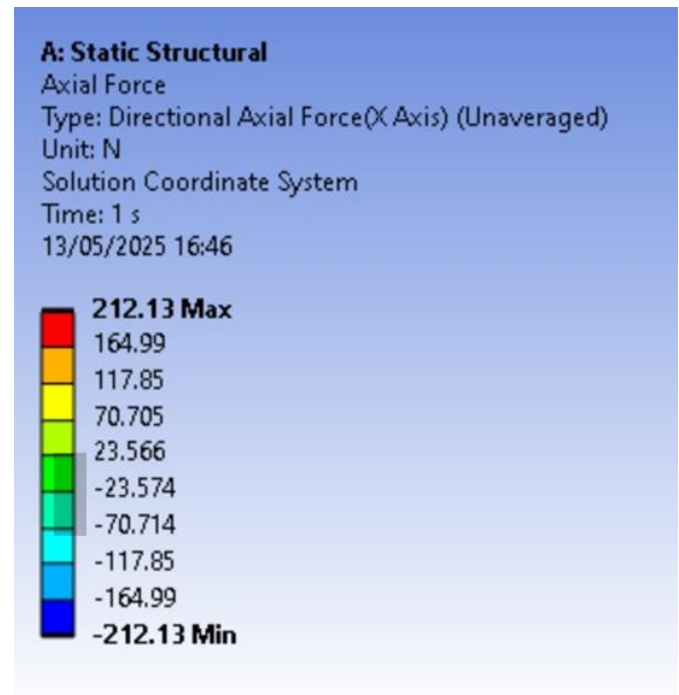
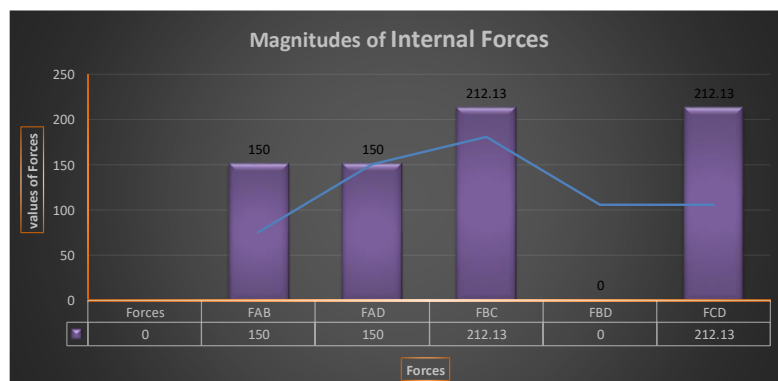
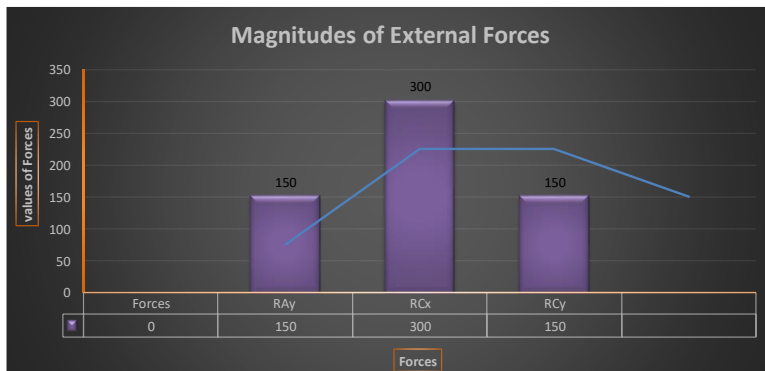
Horizontal:

$$\begin{aligned}F_{AB} + F_{AD} \cos 45^\circ &= 0 \\ F_{AB} &= -150 \text{ N} \\ &= 150 \text{ N (Compression)}\end{aligned}$$



J. Table and Graph

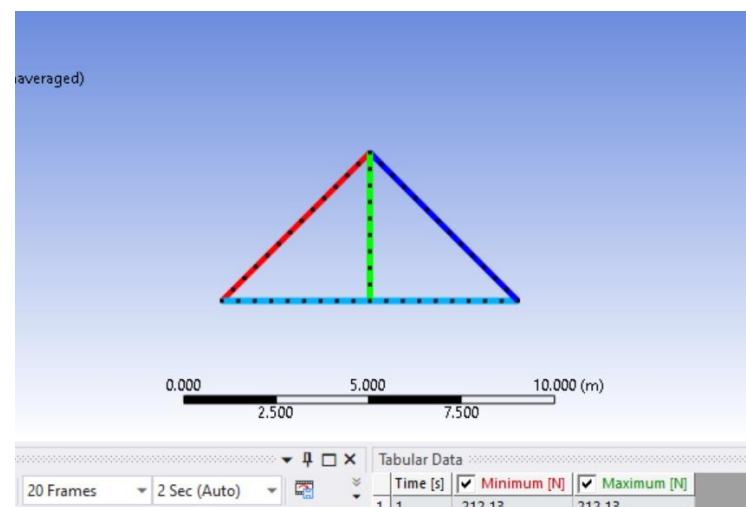
R_{Ay} (N)	R_{Cx} (N)	R_{Cy} (N)	F_{AB} (N)	F_{AD} (N)	F_{BC} (N)	F_{BD} (N)	F_{CD} (N)
150 (Down)	300 (Left)	150 (Up)	150 (C)	212.13 (T)	150 (T)	0 (Zero Force)	212.13 (C)



According to the results from ANSYS:

- The maximum axial force is **212.13 N** (shown in red) in member AD, which matches the calculated tensile force.
- The minimum axial force is **-212.13 N** (shown in dark blue) in member CD, confirming it is in compression as manually calculated.
- Member BD is shown in green, indicating **zero axial force**, which matches our result that it is a **zero-force member**.
- The remaining members (AB and BC) show values of ± 150 N, consistent with the **manual calculation** of compression and tension respectively.

K. Ansys Verification



The close match between simulated and manual results confirms the accuracy of our analytical method and calculations. The axial force distribution shown in the simulation confirms the values calculated earlier.

L. Results and Discussion

After solving the truss using the method of joints, we calculated internal forces for each member and found the following:

- Member **AD**: 212.13 N (Tension)
- Member **AB**: 150 N (Compression)
- Member **BD**: 0 N (Zero-force)
- Member **BC**: 150 N (Tension)
- Member **CD**: 212.13 N (Compression)

The external reactions were also determined:

- $R_{Ax} = 0\text{ N}$
- $R_{Ay} = -150\text{ N}$ (Downward),
- $R_{Cy} = 150\text{ N}$ (Upward),
- $R_{Cx} = 300\text{ N}$ (Leftward).

To verify the accuracy of our hand calculations, we simulated the same truss in ANSYS. The ANSYS results matched our calculations, showing a maximum axial force of **212.13 N** and a minimum of **-212.13 N**, indicating that the directions and magnitudes of member forces were correctly predicted. This close agreement supports the validity of our manual analysis.

M. Conclusion

The task was to analyze a given truss, find internal member forces and support reactions, and verify them. We applied the method of joints to solve the truss manually and confirmed the results using ANSYS simulation. Both methods gave matching values, proving the analysis was correct.

N. References

1. Hibbeler, R. C. *Engineering Mechanics: Statics*, 14th Edition, Pearson, 2016.
2. ANSYS Workbench Documentation, 2025.